

Software Engineering Lab – Lab 3

Component Modelling & Architectural Pattern Selection

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Architecture Selection

We chose Layered Architecture for the Vaccination Cohort & Dose Scheduling System.

Architectural Choice

Layered Architecture separates the system into Presentation, Business, and Data layers. This provides a clear separation between user interaction, vaccination-related business logic, and data storage.

Reason 1

The system contains several functions such as citizen registration, vaccination scheduling, dose interval validation, vaccination center slot management, and QR certificate generation. Separating these responsibilities into appropriate layers makes the system easier to understand and maintain.

Reason 2

The vaccination system must enforce rules such as the minimum interval between vaccination doses before allowing the next dose to be booked. A dedicated business layer can handle these rules while the data layer stores citizen, vaccination, booking, and certificate information.

Security Advantage

The layered architecture prevents users from directly accessing the database. Requests can be validated and authorized through the business layer before sensitive vaccination information is accessed.

Performance Benefit

Separating business processing from data storage allows frequently required operations, such as dose-interval validation and vaccination-slot checking, to be processed efficiently without exposing the database directly to users.

Diagram

